Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Disaster-Monitoring Sensing Networks

CASTER-ZT

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Roadmap

1. Context & Motivations
2. Related Work
3. Problem Statement
4. Research Questions
5. Research Objectives
6. Architecture
7. Methodology
8. Results
9. Progress & Future Work

Context & Motivations | The Autonomous Recovery Challenge

Disaster-Monitoring Sensing Networks

- Large-scale simultaneous failures ($\approx$40% cells offline)
- Degraded observability, rapidly shifting priorities
- Manual recovery is **too slow** for 6G requirements
- AI-native systems must act in **near-real-time** ($<$10 ms) [1]

The Autonomy Dilemma

- Same autonomy that *speeds recovery* also *enlarges the attack surface*
- Corrupted telemetry $\Rightarrow$ poisoned decisions
- Stolen credentials $\Rightarrow$ rogue autonomous actuation [2, 3]

Why 6G Makes This Urgent

- 6G positions AI as a **first-class architectural element** [4]
- O-RAN Near-RT RIC enforces **10 ms actuation budget** [1]
- Edge-only inference — no cloud round-trip
- EU AI Act classifies critical-infrastructure AI as **high-risk**: requires auditable decisions [5]

Graph Learning Fits Networked Recovery

GNNs naturally capture topology [6, 7, 8], but **no existing framework** combines graph decisions with formally bounded zero-trust security.

Attack Landscape

- **Telemetry poisoning:** corrupted sensor readings manipulate policy input
- **Identity-credential abuse:** stolen legitimate credentials inject rogue recovery actions
- **Combined attack:** simultaneous telemetry + identity compromise
- Adversarial attacks on GNNs are well-documented [9, 10]

Critical Observation [3]

"Never trust, always verify" — zero-trust requires continuous **per-action** verification, not just perimeter defense.

State of Safe / Shielded Learning

- Safe RL constrains cumulative cost [11, 12]
- Binary shields use temporal specs [13]
- Anomaly detectors flag suspicious inputs [14]

The Gap

No existing method jointly provides:

- Graph-structured state representation
- Identity-aware zero-trust gating
- Formally bounded, graduated decisions
- Uncertainty-based abstention

Related Work

Family	Objective (resolves what?)	Methodology (to resolve what?)	Key Result	Limitation
GNN Routing [15, 16]	Resolve **scalability of routing & QoS** in large-scale networks with complex topologies	GNN-based policy; supervised learning on topology graphs — to learn network-wide state and compute routing decisions at scale	SoA QoS; low overhead	**No security gating;** assumes benign environment
Safe RL [11, 12]	Resolve **unsafe policy execution**: bound cumulative cost during RL training & deployment	Lagrangian CPO; cost critic — to enforce safety constraints without sacrificing task performance	Constraint satisfaction in robotics	**No identity/trust**; cost cannot capture identity attacks
Formal Shield [13, 17]	Resolve **formal safety violations at execution**: enforce temporal safety specs from system requirements	Precomputed binary shield from LTL/CTL; DMPS separates learned perception from deterministic safety — to guarantee no unsafe action executes	Provably safe actions; zero spec violations	**Binary only;** no graduated decisions; no identity/trust
Anomaly / Trust [14, 18]	Resolve **corrupted/malicious sensor readings**: detect faulty or adversarial inputs before they affect WSN decisions	Isolation Forest, SVM, DBSCAN — to identify and isolate abnormal readings from sensor streams	Good detection; lightweight	**No formal shield;** high FP; no graduated enforcement
Agentic AI / O-RAN [19, 20]	Resolve **orchestration rigidity**: enable flexible adaptive autonomous control in O-RAN	LLM-based / intent-driven xApp; multi-agent orchestration — to support real-time multi-intent negotiation and adaptive control	Flexible orchestration; fast adaptation	**No formal security bound;** no identity verification under attack

Problem Statement

Central Gap

Current methods optimize recovery quality **before** defining a formal admissibility boundary for autonomous action under compromised or uncertain evidence — a **dangerous ordering** in disaster settings.

Formal Decision Space

The shield produces **graduated** decisions ordered by conservatism:

$$\mathcal{D} = \{\textsc{allow},\ \textsc{scope-reduce},\ \textsc{defer},\ \textsc{escalate},\ \textsc{block}\}$$

Optimization Objective

$$\max_{\pi_\theta}\ \mathbb{E}\Big[\sum_t U(a_t^\star, s_t) - \lambda_1 U_{\text{sec}} - \lambda_2 U_{\text{lat}} - \lambda_3 U_{\text{ener}}\Big]$$

$$\text{subject to}\quad a_t^\star \in \mathcal{E}(d_t, \hat{a}_t) \qquad \text{(shield must approve every action)}$$

System Setting

- Network modeled as time-varying directed graph $G_t = (V_t, E_t, X_t)$
- 6 recovery action types: power-cycle, reroute-traffic, reassign-frequency, cell-reconfig, isolate-cell, activate-backup
- Disaster onset at tick 3: $\approx$40% simultaneous cell failure
- Bounded adversary: single zone, 2 severity levels

6G Near-RT Compliance [1]

- Per-decision latency $\leq$10 ms
- Edge-local inference (no cloud)
- Model memory $\leq$50 MB

Parameter Definitions

$G_t=(V_t, E_t, X_t)$: graph (nodes=cells, edges=links, features X_t) · s_t: system state · π_θ: learned policy · $\hat{a}_t$: proposed action · $a_t^\star$: executed action · $\mathcal{E}(d_t, \hat{a}_t)$: admissible set given shield decision d_t · $U(a_t^\star, s_t)$: recovery utility · $\lambda_1, \lambda_2, \lambda_3$: penalty weights for U_{sec} (security), U_{lat} (latency), U_{ener} (energy)

Research Questions

The article states four research questions (no overarching main question is formulated in the paper).

RQ1 — Security Effectiveness

Can a zero-trust shielded policy **reduce adversarial degradation** of autonomous recovery?

RQ2 — Security–Utility Tradeoffs

What **security–utility–overhead tradeoffs** arise from trust-, authorization-, and risk-aware shielding?

RQ3 — Generalization

Does the framework remain **effective across attack severities and topology scales**?

RQ4 — Deployment Compliance

Does the model satisfy **6G near-real-time deployment constraints**?

Research Objectives

One objective per research question (contributions C1–C5 from the article are mapped below).

O1 (answers RQ1) — Formulate & Build the Shielded Framework

Formulate secure autonomous recovery as a **graph-structured decision problem** with zero-trust admissibility constraints [C1], and introduce CASTER-ZT coupling GNN encoder, learned policy, trust–risk evaluator, MC-Dropout uncertainty, and a **formally verified deterministic shield** [C2].

O2 (answers RQ2) — Establish Formal Guarantees & Ablate Components

Prove **10 theoretical properties** (defense-in-depth, conformal coverage, calibration convergence) [C3] and confirm each shield module's individual necessity via component ablation [C5].

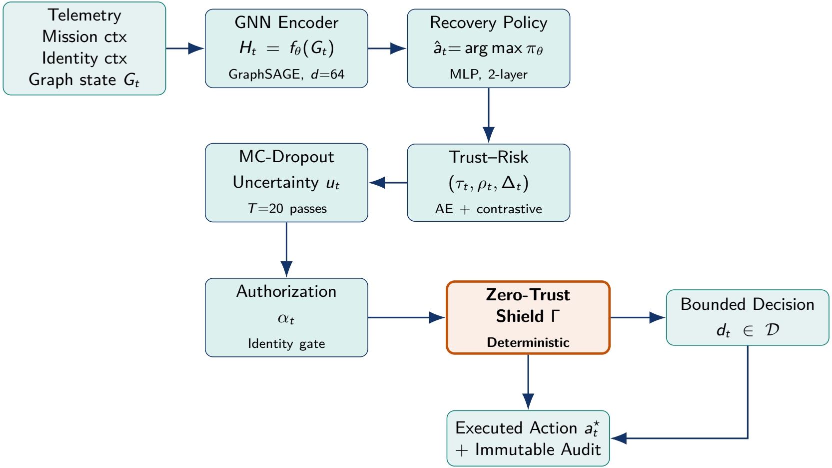
O3 (answers RQ3) — Validate Generalization Across Scales & Attacks

Run a **2,420-run campaign** across 3 topology scales, 4 attack conditions, 4 external baselines and 20 seeds to demonstrate 74.2–98.3% rogue detection with zero false positives [C4].

O4 (answers RQ4) — Verify 6G Near-RT Deployment Compliance

Demonstrate $\leq$3.07 ms per-decision latency, $\leq$2 MB model memory, and edge-local autonomy — satisfying the O-RAN Near-RT RIC 10 ms budget [C4].

Architecture | CASTER-ZT End-to-End Dataflow



Design principle: **The policy proposes; the shield decides admissibility.** AI-native coverage $\eta_{\mathrm{AI}} = 5/6$.

Architecture | Five-Layer Component Breakdown

#	Component	Architecture	Role	Parameters
1	GNN Encoder f_θ *(learned)*	2-layer GraphSAGE [7], mean aggregation, d=64, ReLU	Encodes graph topology into node and graph-level embeddings	≈4.5K
2	Recovery Policy π_θ *(learned)*	2-layer MLP: $[h_a \| \bar{h}_{G_t}] \rightarrow$ action logits, softmax	Proposes the best candidate recovery action $\hat{a}_t$	≈8.5K
3	Trust–Risk Evaluator Q_ϕ *(learned)*	Autoencoder (trust τ_t) + Contrastive encoder (divergence Δ_t) + Neural risk scorer (ρ_t)	Dual-hypothesis safety assessment under benign vs. adversarial hypothesis	≈6.9K
4	MC-Dropout U_ω *(learned)*	T=20 stochastic forward passes; variance over action distribution	Epistemic uncertainty estimation [21]; triggers escalation or block when u_t is high	p=0.1
5	Zero-Trust Shield Γ *(deterministic)*	14 calibrated scalar thresholds; formal admissibility gate (Algorithm 1)	Sole decision authority; enforces all 5 outcomes; formally verified [3]	14 scalars
	Total			≈ **20K** + shield

Notation: f_θ: GNN encoder; π_θ: recovery policy; Q_ϕ: trust–risk evaluator; τ_t: trust score; ρ_t: risk score; Δ_t: hypothesis divergence; U_ω: MC-Dropout uncertainty module; u_t: epistemic uncertainty; Γ: zero-trust shield; d_t: shield decision; d: embedding dim.; p: dropout rate. Model memory ≤2 MB — edge-deployable, no GPU at inference.

Methodology | Zero-Trust Shield Algorithm

Shield Decision Logic (Algorithm 1)

1. Encode: $H_t \leftarrow f_\theta(G_t)$
2. Policy: $\hat{a}_t \leftarrow \arg\max \pi_\theta(a|s_t)$
3. Authorization: $\alpha_t \leftarrow A(i_t, \hat{a}_t, s_t)$
4. Trust: $\tau_t \leftarrow T_\phi(s_t)$ (AE recon. error)
5. Risk: $\rho_t \leftarrow R_\phi(\hat{a}_t, s_t)$
6. Divergence: $\Delta_t = \|q_\phi^{(0)} - q_\phi^{(1)}\|_2$
7. Uncertainty: $u_t \leftarrow U_\omega(s_t, \hat{a}_t)$
8. Conservatism: $g_t = w_1(1-\tau_t) + w_2\rho_t + w_3 u_t + w_4\Delta_t$

Graduated Output Rule

- $\alpha_t{=}0$ or $\Delta_t{>}\delta_{\max}^{\text{hard}} \Rightarrow$ **block**
- $g_t{<}\gamma_1$ and $\tau_t{\geq}\tau_{\min} \Rightarrow$ **allow**
- $\gamma_1{\leq}g_t{<}\gamma_2 \Rightarrow$ **scope-reduce**
- $\gamma_2{\leq}g_t{<}\gamma_3 \Rightarrow$ **defer**
- $\gamma_3{\leq}g_t{<}\gamma_4$ or $u_t{>}u_{\max} \Rightarrow$ **escalate**
- else $\Rightarrow$ **block**

Training Pipeline

- 2,000 simulated episodes; ≈60K annotated steps
- 70/15/15 train/val/test split
- All 4 attack conditions + clean

Component	Loss	Epochs
GNN + Policy	CE	15–20
Trust AE	MSE	10–20
Risk scorer	BCE	25–30
Contrastive enc.	Margin	40–50
Shield Γ	Calibration	—

Complexity

Per-decision: $\mathcal{O}(L|E_t|d + |\mathcal{A}|k)$
Threshold calibration: $\mathcal{O}_P(1/\sqrt{n})$ convergence

Safety Properties

P1 **Monotone conservatism:** worsening any signal ($\uparrow \rho_t$, $\downarrow \tau_t$, $\uparrow u_t$, $\uparrow \Delta_t$) cannot produce a less conservative decision [13]

P2 **Unauthorized-actuation exclusion:** $\alpha_t{=}0 \Rightarrow d_t{=}\textsc{block}$ (always)

P3 **Divergence safeguard:** $\Delta_t{>}\delta^{\text{hard}}_{\text{max}} \Rightarrow d_t{=}\textsc{block}$

P4 **Bounded completeness:** Γ is a total function — every input maps to exactly one $d_t \in \mathcal{D}$

P5 **Complexity bound:** $\mathcal{O}(L|E_t|d + |\mathcal{A}|k)$

P6 **Defense-in-depth:** $\Pr[\textsc{allow}|\text{attack}] \leq \varepsilon_\alpha \cdot \varepsilon_\tau$ (independent gates multiply)

Statistical & Utility Properties

P7 **Calibration accuracy:**
$\Pr[\varepsilon^\star \leq \hat{\varepsilon} + \sqrt{\ln(1/\delta)/(2n)}] \geq 1-\delta$ (Hoeffding) [22]

P8 **Bounded price of safety:**
$\mathbb{E}[\text{utility loss}] \leq \eta \cdot U_{\text{gap}} + \eta_{\text{block}} \cdot U_{\text{max}}$

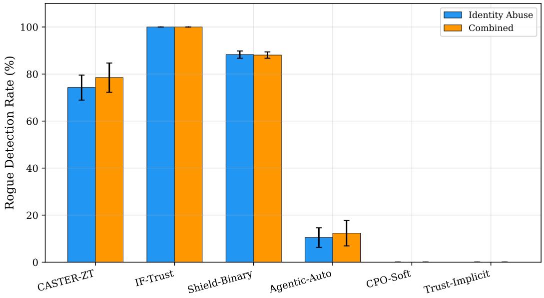
P9 **Conformal coverage:** distribution-free guarantee that unsafe actions are blocked with probability $\geq 1{-}\alpha$ [22]

P10 **Calibration convergence:** $|\gamma_1^{(n)} - \gamma_1^\star| = \mathcal{O}_P(1/\sqrt{n})$

Key Insight

The deterministic shield is **policy-agnostic**: any future controller (RL agent, LLM-based planner [19]) can serve as proposal generator without modifying the shield.

Results | RQ1 — Security Effectiveness Under Attack



Rogue detection and blocking under high-severity identity abuse. CASTER-ZT: 74.2% detection, ***zero false positives****; Shield-Binary: 88.3% with 48.8% FP.*

Key Results — High-Severity Identity Abuse (20 seeds)

Method	RogueDet	FalseBlk	ω_{rec}
CASTER-ZT	**.742**	**.000**	**.733**
IF-Trust [14]	1.000	.000	.925
Shield-Bin. [13]	.883	.488	.758
Agentic [19]	.105	.568	.144
CPO-Soft [11]	.000	.000	.201

Differentiator

CASTER-ZT is the only method achieving zero false positives. IF-Trust achieves 100% detection but throttles ALL actions — even under clean conditions (pathological). Shield-Binary blocks 48.8% of legitimate actions.

Results | RQ2 — Recovery Quality & Ablation

Recovery Quality — High-Severity Identity Abuse

Method	ω_{rec}	p-value
CASTER-ZT	**0.733 ± 0.038**	—
IF-Trust	0.925 ± 0.000	<0.001
Shield-Binary	0.758 ± 0.013	0.064
Agentic-Auto	0.144 ± 0.027	<0.001
CPO-Soft	0.201 ± 0.030	<0.001
Trust-Implicit	0.201 ± 0.030	<0.001

Non-shielding baselines achieve only 0.14–0.20 recovery quality ($p<0.001$).

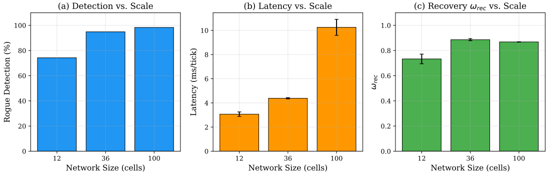
Ablation Study — What Each Gate Contributes

Ablation	RogueDet	ω_{rec}
Full CASTER-ZT	**.742**	**.733**
−Trust	.524	.380
−Risk	.665	.678
−Authorization	.763	.760

Finding

Trust assessment is the critical gate: removing it causes RogueDet to drop from 0.742 to 0.524 and ω_{rec} from 0.733 to 0.380 ($p<0.001$).
Validates Proposition 6: independent gates multiply to reduce unsafe execution probability.

Results | RQ3 — Generalization Across Scales & Severities



Security improves with scale; 12-cell latency is 3× below O-RAN 10 ms budget.

Multi-Scale Results — High-Severity Identity Abuse

Topology	RogueDet	ω_{rec}	Latency
12-cell, 2-zone	0.742	0.733	3.07 ms
36-cell, 4-zone	0.949	0.886	4.38 ms
100-cell, 8-zone	0.983	0.868	10.25 ms

Key Observations

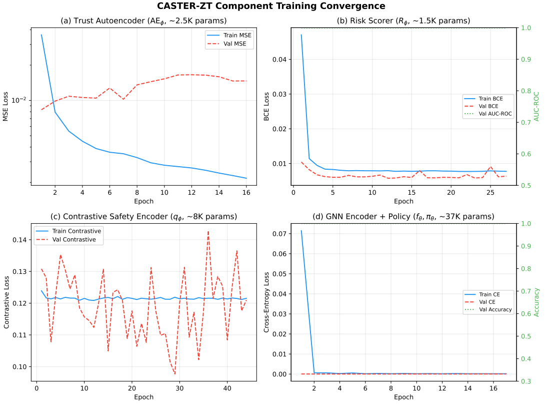
- Detection **improves with scale** (richer structural context for GNN)
- Latency scales **within O-RAN budget** at 12 and 36 cells
- Zero false positives **across all scales** and all $\gamma_1 \in \{0.10, \ldots, 0.60\}$ (threshold sensitivity confirmed)
- Severity: RogueDet 0.806 (medium) $\rightarrow$ 0.742 (high) — consistent degradation

Deployment Profile

Metric	Value	Limit
Latency (12-cell)	3.07 ms	<10 ✓
Latency (36-cell)	4.38 ms	<10 ✓
Latency (100-cell)	10.25 ms	<10 ○
Model memory	≤2 MB	<50 ✓
Parameters	≈ 20K	—
AI-native η_{AI}	5/6	— ✓
Audit completeness	100%	Req. ✓
Edge autonomy	Yes	Req. ✓

100-cell note

10.25 ms marginally exceeds O-RAN 10 ms budget.
Future work: quantization, edge batching.



Training convergence for all 4 learned components (≈20,019 params). Trust AE: MSE by epoch 16. Risk scorer: BCE, AUC= 0.999 by epoch 27. Contrastive encoder: margin by epoch 44. GNN+Policy: CE, acc= 1.000 by epoch 17.

Progress & Future Work

✓ Completed Work

- **Framework:** Full CASTER-ZT pipeline designed and implemented (GNN encoder, policy, trust–risk evaluator, MC-Dropout, shield)
- **Theory:** 10 formal propositions proved (supplemental proofs document)
- **Evaluation:** 2,420-run campaign (11 methods $\times$ 7 conditions $\times$ 20 seeds + multi-scale)
- **Baselines:** 4 external baselines + 6 ablations faithfully implemented
- **Data:** 4-layer data provenance (synthetic + 5G real distributions + SWaT/WADI grounding + conformal calibration)
- **Compliance:** 6G near-RT, O-RAN, EU AI Act requirements analyzed and met at primary scale
- **Paper:** Manuscript drafted for IEEE Transactions on Mobile Computing (TMC) — *(Submitted)*

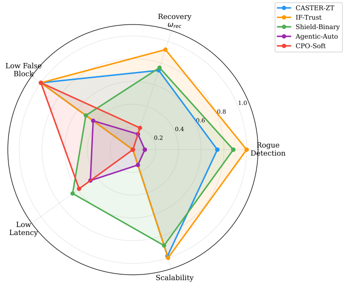
○ Limitations & Future Work

- **Trust AE vs. telemetry poisoning:** reconstruction-error detector does not hard-block sophisticated poisoning attacks [23, 24]
  $\hookrightarrow$ *Plan: cross-source GNN consistency checking, adversarially trained detectors, temporal sequence modeling*
- **100-cell latency:** 10.25 ms marginally exceeds O-RAN 10 ms budget
  $\hookrightarrow$ *Plan: model quantization, edge batching, hardware profiling*
- **Policy learning:** current policy trained via supervised imitation
  $\hookrightarrow$ *Plan: RL-based online policy optimization under shield constraints*
- **Adaptive adversaries:** adversary cannot currently modify shield internals
  $\hookrightarrow$ *Plan: evaluation under adaptive attackers, multi-zone scenarios*
- **Agentic integration:** combine with LLM-based orchestrator as proposal generator [19, 25]

Conclusion

Summary

- Proposed **CASTER-ZT**: first framework combining graph-structured decision making with formally bounded zero-trust admissibility for disaster-recovery autonomy
- **10 formal properties**: defense-in-depth, conformal coverage, calibration convergence
- **Zero false positives** across all conditions — unique among all evaluated methods
- **3.07 ms** per-decision latency — within O-RAN Near-RT RIC budget
- $\approx$**20K params,** $\leq$**2 MB** — fully edge-deployable, no GPU
- Policy-agnostic shield: future-proof for RL or LLM-based proposal generators



Radar: CASTER-ZT unique zero-FP, graduated-enforcement profile under high-severity identity abuse.

Bottom Line

Security-first $\neq$ utility-last. CASTER-ZT proves you can enforce formal zero-trust boundaries *and* deliver meaningful autonomous recovery.

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